

Computer Networks — Midterm Syllabus (Lectures 1–13)

CSAI Computer Networks (AY 26-27, Sem-5) | Weeks 1–8

Week 1–2: How the Internet Works

Lecture 1 — Course Introduction & the Instagram Journey

- Internet as a "network of networks"; nuts-and-bolts view vs. service view
- Network **edge** (end systems, access networks: WiFi, cellular, Ethernet) vs. **network core** (routers/switches)
- End-to-end journey of a request: phone → access network → ISP → core → data centre/CDN
- **traceroute** basics; latency by hop (access vs. core vs. server)
- Cloud mirror: AWS Region, Availability Zone (AZ), Edge Location (vocabulary only)

Lecture 2 — Packet Switching & Delay, Loss, Throughput

- Packet switching (store-and-forward, statistical multiplexing) vs. circuit switching
- Four delay components: **processing, queuing, transmission, propagation**
- Transmission delay = L/R ; Propagation delay = d/s (plug-in calculations only, no queuing math)
- Packet loss (buffer overflow, conceptual only)
- Throughput and the **bottleneck link** concept

Lecture 3 — Protocol Layers: OSI vs. TCP/IP

- Purpose of layering: service, interface, protocol; modularity/interoperability
- OSI 7-layer model (mnemonic: All People Seem To Need Data Processing) vs. TCP/IP 5-layer model
- Layers and PDUs: Application (message) → Transport (segment) → Network (datagram) → Link (frame) → Physical (bits)
- Encapsulation / decapsulation
- Which device operates at which layer: host = L7, router = L3, switch = L2
- AWS layer mapping: Route 53/ALB (L7), NLB (L4), VPC route tables/IGW (L3)

Lecture 4 — Network Devices, Topologies, and AWS VPC

- Devices by layer: Hub (L1), Switch (L2/MAC), Router (L3/IP), Firewall (L3–L7), Load Balancer (L4/L7)
 - Collision domains vs. broadcast domains (light touch)
 - Topologies: Star, Mesh, Hybrid/tiered — trade-offs of cost vs. resilience
 - AWS cloud network: **Region**, **Availability Zone (AZ)**, **Edge Location**
 - VPC = software-defined private network; default VPC components (subnets, route table, Internet Gateway)
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Week 3–4: The Application Layer

Lecture 5 — HTTP/1.1 → HTTP/2 → HTTP/3, Client–Server vs. P2P, REST

- Client–server vs. peer-to-peer (P2P) architectures
- HTTP request anatomy (method, URL, headers, body) and response (status line, headers, body)
- Methods: GET, POST, PUT, DELETE; status classes 2xx/3xx/4xx/5xx
- Persistent vs. non-persistent connections; **head-of-line (HoL) blocking**
- HTTP/1.1 (persistent, ~6 parallel connections) → HTTP/2 (multiplexing over one TCP connection, HPACK, still has TCP-level HoL blocking) → HTTP/3 (runs over **QUIC/UDP**, independent streams, 0-1-RTT setup)
- REST principles: resources, HTTP methods as CRUD, statelessness

Lecture 6 — HTTPS and the TLS Handshake

- HTTPS = HTTP over TLS; three goals: **confidentiality, integrity, authentication**
- Symmetric vs. asymmetric encryption; why TLS uses asymmetric to bootstrap a symmetric session key
- Certificates: domain ↔ public key, signed by a CA; chain of trust (leaf → intermediate → root)
- TLS handshake steps: ClientHello → ServerHello+Certificate → Authentication → Key exchange → Finished (symmetric encryption begins)
- TLS 1.3 ≈ 1-RTT handshake; link to HTTP/3's fast setup
- Man-in-the-middle (MITM) defeated by certificate validation
- AWS Certificate Manager (ACM) — automates cert issuance, attached at a load balancer (TLS termination)

Lecture 7 — DNS: The Internet's Phone Book, and AWS Route 53

(Note: numbered "Lecture 8" in the source outline, delivered as Week 4's first lecture)

- DNS translates names → IP addresses; runs before TCP/TLS
- Hierarchy: Root → TLD (.com) → Authoritative name servers
- **Recursive** resolution (resolver does the legwork) vs. **iterative** queries (referrals)
- Record types: A (IPv4), AAAA (IPv6), CNAME (alias), MX (mail), NS (name servers), TXT
- Caching and **TTL**; why DNS changes propagate slowly
- AWS Route 53 routing policies: **Simple, Weighted, Latency-based, Failover, Geolocation**

Lecture 8 — Email Protocols (SMTP, POP3, IMAP), CDNs and Caching

(Note: numbered "Lecture 7" in the source outline, delivered as Week 4's second lecture)

- Web caching: cache hit/miss; **Cache-Control**, **Etag**/If-None-Match → 304 Not Modified
 - CDN: distributed edge servers caching content near users; pull vs. push caching; DNS-based steering to nearest edge
 - AWS CloudFront + Edge Locations
 - Email architecture: Mail User Agent ↔ Mail Transfer Agents ↔ mailboxes
 - **SMTP** (push/send, port 25/587) vs. **POP3** (download-and-delete) vs. **IMAP** (server-synced, multi-device)
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Week 5–6: The Transport Layer & Cloud Load Balancing

Lecture 9 — Transport Layer: Multiplexing, Ports, UDP, and the TCP Connection

- Transport layer = process-to-process delivery via **ports** (vs. network layer's host-to-host via IPs)
- Multiplexing/demultiplexing; a TCP connection identified by the **4-tuple** (src IP, src port, dst IP, dst port)
- Well-known ports: 22 (SSH), 25/587 (SMTP), 53 (DNS), 80 (HTTP), 443 (HTTPS)
- **UDP**: connectionless, no handshake/ACKs/ordering — used for DNS, live video, VoIP, QUIC/HTTP-3
- **TCP**: connection-oriented, reliable, in-order byte stream
- TCP **3-way handshake**: SYN → SYN-ACK → ACK; teardown via FIN/ACK
- Choosing UDP vs. TCP based on reliability/latency needs

Lecture 10 — Reliable Data Transfer (Go-Back-N, Selective Repeat) and TCP Flow Control

- Building reliability: checksums, ACK/NAK, timeouts, retransmission, sequence numbers
- Stop-and-wait (correct but slow) → pipelining (windows) for utilization
- **Go-Back-N (GBN)**: cumulative ACKs, resends entire window tail from lost packet onward
- **Selective Repeat (SR)**: individual ACKs, resends only the lost packet, receiver buffers out-of-order packets
- Real TCP is a hybrid (cumulative ACKs + SACK + fast retransmit)
- TCP sliding window; **flow control** via receiver-advertised window (**rwnd**) — protects the *receiver*

Lecture 11 — TCP Congestion Control (AIMD, Slow Start) and TLS at the Transport Edge

- Congestion = offered load > network capacity → queuing delay and loss
- **Congestion window (cwnd)** — protects the *network* (vs. *rwnd* protecting the receiver); sender rate $\approx \min(\text{cwnd}, \text{rwnd})$
- **Slow start**: cwnd grows exponentially until ssthresh or loss
- **Congestion avoidance**: cwnd grows linearly (additive) beyond ssthresh
- **AIMD** (Additive Increase, Multiplicative Decrease): +1 MSS/RTT increase, $\times \frac{1}{2}$ decrease on loss → fairness; produces the classic **sawtooth**
- Loss response: timeout → full reset to slow start; triple duplicate ACK → fast recovery (halve cwnd)
- TLS rides on top of TCP's reliable byte stream (HTTPS = HTTP over TLS over TCP); QUIC re-implements reliability over UDP for HTTP/3

Lecture 12 — Cloud Load Balancing: ALB vs. NLB, Elastic Load Balancing, High Availability

- Load balancer goals: scalability, availability, manageability
 - **NLB (Layer 4)**: forwards by IP/port (TCP/UDP), high throughput, no HTTP awareness
 - **ALB (Layer 7)**: HTTP-aware, routes by host/path, terminates TLS
 - Target groups and **health checks** — automatic removal/re-addition of unhealthy targets
 - **Auto Scaling Group (ASG)** for elasticity; **multi-AZ** deployment for high availability (no single point of failure)
 - End-to-end request trace (capstone): Route 53 → TCP handshake → TLS/ACM → HTTP routing (ALB) → healthy target → congestion-controlled response
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Week 7–8: Addressing, Subnetting, NAT, and Diagnostics

Lecture 13 — IPv4 Addressing, Classful History, and CIDR

- IPv4 address: 32 bits, dotted-decimal, split into network portion + host portion
 - Classful addressing (A/B/C) and why it was abandoned (wasted address space)
 - **CIDR** notation a.b.c.d/n; subnet mask as dotted-decimal form of the prefix
 - **Usable hosts** = $2^{(32-n)} - 2$; sizing a prefix for a host requirement (round up to next power of 2)
 - Private address ranges (RFC 1918): 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16
 - AWS VPC CIDR blocks use the same addressing math
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Quick-Reference: Key Formulas & Facts

Concept	Formula / Fact
Transmission delay	L / R (packet size ÷ link rate)
Propagation delay	d / s (distance ÷ signal speed, $\approx 2 \times 10^8$ m/s in fibre)
Usable IPv4 hosts	$2^{(32-n)} - 2$
AWS usable hosts	$2^{(32-n)} - 5$
Subnets from borrowed bits	2^k subnets from borrowing k host bits
TCP handshake	SYN → SYN-ACK → ACK
DHCP sequence	Discover → Offer → Request → Ack (DORA)
Private IPv4 ranges	10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16
AIMD	Additive Increase (+1 MSS/RTT) / Multiplicative Decrease ($\times \frac{1}{2}$ on loss)
Sender rate	$\approx \min(\text{cwnd}, \text{rwnd}) / \text{RTT}$

OSI / TCP-IP Layer Cheat Sheet

Layer (TCP/IP)	PDU	Example Protocols	Device
Application	Message	HTTP, DNS, SMTP	Host
Transport	Segment	TCP, UDP	Host
Network	Datagram	IP	Router
Link	Frame	Ethernet, WiFi	Switch
Physical	Bits	—	Hub/cabling
